

Project Overview

Aims of the Project:

- The aim of the MCTR1021 course project is to generate optimal or near-optimal solutions for real-life problems related to **multi-agent cooperative systems** via meta-heuristic optimization techniques.
- Students can choose projects under the umbrella of:
Multi-agent cooperative systems in robotics and/or autonomous systems (e.g. autonomous vehicles or UAVs, ... etc)

Hardware:

- No hardware is required in this project context.

Programming Languages Allowed:

- You can choose either **MATLAB or Python** to implement your projects.
- All techniques, visualizations and simulations will be implemented using the chosen language.
- The chosen language **will not be changed** after the implementation of the first algorithm.

Teams:

The team formation regulations are:

- Each team will be composed of **4-6 members**.
- No exceptions will be made to form a team less than or more than the specified number of team members mentioned above.
- Students not assigned to teams by the deadline of teams registration will be randomly clustered or assigned into groups and added to the list of teams.
- No change to the project teams (interchange to another team or exchange of project) will be accepted under any circumstances.

Project Selection:

- Each team will be asked to review **5 recent scientific articles total** (i.e. publication date between 2019-2026) related to the previously mentioned themes (There will be a file uploaded in the CMS that includes previous publications from the course).
- From the reviewed papers, each team will select **2 different ideas** from which they will submit project proposals.(note that the 1st idea we be considered as the first priority)
- The proposal should be submitted in the **template** provided on CMS.
- The proposal should be **maximum 2 pages**.
- The course team will review both proposed ideas and one of them will be selected and allocated to the team.

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Submission Deadline:

- The deadline of proposals submission is on **Thursday, 1 October 2026, at 23:59**.

Project Milestones:

- The overall maximum grade for the project is **40% from the course grade**.
- The project is split into **6 milestones** to facilitate and guide you in the implementation.

In the following table, the weight of each of the milestones will be presented.

Milestone	Brief Description	Deadline
1: Proposal (3%)	Two-page proposal of 2 project ideas within the scope of the previously mentioned themes.	Week 4: Thursday, 1 October 2026
2: Formulation (5%)	Two-page literature survey of minimum 5 relevant research articles in the 6 preceding years (2018 – 2026) and Problem Formulation of the assigned project as code.	Week 5: Wednesday, 8 October 2026
3: SA (6%)	Implementation of Simulated Annealing (SA) algorithm on your problem.	Week 7: Monday, 19 October 2026
4: GA (8%)	Implementation of Genetic Algorithm (GA) on your problem.	Week 10: Wednesday, 11 November 2026
5: PSO / ACO/ WOA (8%)	Implementation of either Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Whale Optimization Algorithm (WOA) or Grey Wolf Optimization Algorithm (GWO) on your problem.	Week 13: Wednesday, 2 December 2026
6: New Algorithm + Final Report (10%)	Implementation of a new meta-heuristic population-based optimization algorithm (not taught in the course) on your problem. Final Project Report (6-page IEEE paper format – LaTeX). Poster presenting the project summary Optional Bonus Task: Implementation of machine learning techniques on your problem.	Week 15: Thursday, 17 December 2026

*The weight of each milestone is presented between brackets beside the name of each milestone.

Each milestone will be graded based on several criteria including **but not limited to**:

- Quality of all submitted material.
- For codes implementation: scalability, modularity, documentation (i.e. well commented), etc.
- For results analysis and discussion: analysis of all results (numerical, graphical, performance metrics, parametric studies, statistical studies, etc.) in light of the specific problem on-hand; as well as commenting on noticed trends and ability to highlight recommendations resulting from the analysis and discussion of results.
- For reporting: clarity of presentation and professional discussion with multiple aids (e.g. block diagrams, flowcharts, figures, tables, graphs, charts, etc.)